

CV of the researcher

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Research profile & major achievements

I am an extragalactic astronomer working at the interface of **galaxy formation and evolution, the galaxy–halo connection, AGN triggering & feedback, and weak gravitational lensing**. My work combines statistical probes (**lensing, two-point clustering, satellite kinematics**) to derive causal, quantitative constraints on how galaxies and super massive black hole co-evolve within dark matter haloes.

- **First-author publications** in Nature (2025), ApJ (2024), A&A (2024, 2022, 2021), spanning dwarf-galaxy clustering, halo-mass proxies, AGN hosts and triggering mechanisms, and weak-lensing systematics.
- **Methods & pipelines:** PSF–systematics quantification for weak lensing (UNIONS), joint lensing–clustering–satellite-kinematics inference, image coaddition; reproducible workflows.
- **Current direction:** pushing to cosmic noon ($1 \lesssim z \lesssim 3$) with JWST/DESI (clustering, galaxy-halo connection, AGN/host diagnostics, quenching), and planning VLA observations of dwarf rotation curves to constrain SIDM cross-sections.
- **Open science:** all papers on arXiv; analysis codes on GitHub, to be archived with DOI (MIT license).
- **Community:** invited talks in Japan/China/online; public outreach (USTC Science Week, 2025).

Professional experience

- Postdoctoral Researcher, USTC, Hefei, China — 01/12/2024–present.
- Visiting Doctoral Scholar (Cosmology), CEA Paris-Saclay (Irfu), France — 15/04/2023–31/03/2024 (Host: Dr. Kilbinger, M.) Co-led PSF–density cross-correlation analysis for weak lensing (UNIONS); contributed to intrinsic alignment and black hole–halo mass lensing studies; developed reproducible pipeline.
- Research on galaxy evolution, galaxy–halo connection and AGN–galaxy co-evolution; high- z extensions with JWST/DESI; collaboration on VLA observation for SIDM tests, China — 01/09/2018–present.

Education

- PhD in Astronomy, USTC, Hefei — 01/09/2018–03/12/2024
Dissertation: The influence and constraints of large-scale structure on galaxy evolution
Supervisor: Prof. Wang, H. · Joint training at CEA Paris-Saclay — 15/04/2023–31/03/2024.
- BSc in Applied Physics, Anhui University of Science and Technology, Huainan — 01/09/2014–01/06/2018.

Selected peer-reviewed publications

Published

1. ZHANG, Z., Chen, Y., Rong, Y., Wang, H., Mo, H., Luo, X., Li, H. (2025). *Unexpected clustering pattern in dwarf galaxies challenges formation models*. Nature 642, 47–52. arXiv:2504.03305.
Contribution: First author; clustering measurement, modeling, interpretation.
Significance: Detects unexpectedly strong clustering in dwarf galaxies, challenging standard formation models of dwarfs and dark matter halos.
2. Hervas Peters, F., Kilbinger, M., Paviot, R., Baumont, L., Russier, E., ZHANG, Z., ... Wittje, A. (2025). *UNIONS: a direct measurement of intrinsic alignment with BOSS/eBOSS spectroscopy*. A&A 699, A201. arXiv:2412.01790.
Contribution: Weak lensing analysis.

3. Ma, L., **ZHANG, Z.**, Wang, H., Wu, X. (2025). *Measuring Mass of Gas in Central Galaxies using weak lensing and satellite kinematics in MOND*. ApJ 984, 101. arXiv:2409.13329.
Contribution: Mentored Ma, L. in the analysis of weak lensing and satellite kinematics.
4. **ZHANG, Z.**, Kilbinger, M., Hervas Peters, F., Li, Q., Luo, W., Baumont, L., Cuillandre, J.-C., Fabbro, S., Gwyn, S., McConnachie, A., Wittje, A. (2024). *Point spread function errors for weak lensing – density cross-correlations: Application to UNIONS*. A&A 691, A75. arXiv:2405.03434.
Contribution: First author, lead analysis & interpretation, methodology & validation.
Significance: Quantifies PSF-induced systematics in weak lensing, enables more precise shear calibration.
5. Li, Q., Kilbinger, M., Luo, W., Wang, K., Wang, H., Wittje, A., Hildebrandt, H., Van Waerbeke, L., Hudson, M., Farrens, S., Liaudat, T., Liu, H., **ZHANG, Z.**, ... (2024). *Black Hole–Halo Mass Relation from UNIONS Weak Lensing*. ApJL 969, L25. arXiv:2402.10740.
Contribution: Weak lensing analysis.
6. **ZHANG, Z.**, Wang, H., Luo, W., Mo, H., Zhang, J., Yang, X., Li, H., Li, Q. (2024). *Halo Mass-observable Proxy Scaling Relations and Their Dependencies on Galaxy and Group Properties*. ApJ 960, 71. arXiv:2305.06803.
Contribution: First author, lead analysis & interpretation, methodology & validation, joint lensing–satellite-kinematics inference.
Significance: Systematically quantifies the galaxy-halo connection through weak lensing measurements.
7. Hong, H., Wang, H., Mo, H., **ZHANG, Z.**, ... (2023). *Dynamical Hotness, Star Formation Quenching, and Growth of Supermassive Black Holes*. ApJ 954, 183. arXiv:2305.02910.
Contribution: Analysis of galaxy-quenching correlations.
8. **ZHANG, Z.**, Wang, H., Luo, W., Mo, H. J., Jing, Y., Yang, X., Li, H. (2022). *Massive star-forming galaxies have converted most of their halo gas into stars*. A&A 663, A85. arXiv:2112.04777.
Contribution: First author, lead analysis & interpretation.
Significance: Evidence for high baryon-conversion efficiency in massive star-forming galaxies.
9. **ZHANG, Z.**, Wang, H., Luo, W., Mo, H. J., Liang, Z., Li, R., Yang, X., ... Shi, J. (2021). *Hosts and triggers of AGNs in the Local Universe*. A&A 650, A155. arXiv:2012.10640.
Contribution: First author, lead analysis & interpretation.
Significance: Quantifies the triggering of type II AGNs in the local Universe.

Submitted / in review

10. Liu, H., Lu, Y., Hui, H., Wang, H., Mo, H., Wang, J., Ouyang, W., **ZHANG, Z.**, Enci Wang, ... (2025). *AI reveals efficacy of radio-AGN quenching only in a minority of galaxies*. under review; preprint forthcoming.
Contribution: Weak lensing analysis.

Open/scientific outputs

- Dwarf clustering code (Nature 21/05/2025): https://github.com/ChenYangyao/dwarf_assembly_bias
- UNIONS PSF–systematics code (A&A 30/10/2024): https://github.com/ziwenkk/Lambda_statistics
- Image coaddition pipeline (01/09/2020–10/05/2021), core development.

Invited talks

- IPMU APEC Seminar, Chiba, Japan — 19/06/2025
— Title: Unexpected clustering pattern in dwarf galaxies challenges formation models.
- KIAA–DoA Seminar, Beijing (online) — 23/05/2023
— Title: Halo Mass-observable Proxy Relations and Their Dependencies on Galaxy and Group Properties.

- Xiamen Univ. Haiyun Doctoral Forum, Xiamen, China — 27/07/2022
— Title: Hosts and triggers of AGNs in the Local Universe.

Selected conference presentations & outreach

- Dark Matter & Galaxies Symposium, Tunxi, China — 06/07/2025
— Talk. Title: Unexpected clustering pattern in dwarf galaxies challenges formation models.
- 27th Guo Shoujing Symposium, Changsha, China — 23/05/2025
— Talk. Title: Galaxy—Halo connections.
- USTC Science Week, Hefei, China — 17/05/2025
— Public lecture. Title: Large-Scale Structure of the Universe.
- Science lecture for middle school students, Hefei, China — 13/09/2024
— Public lecture. Title: Stories of Black holes, Galaxies and Dark matter.
- CSST Annual Science Conference, Zhejiang, China — 13/05/2024
— Talk. Title: Halo Mass-observable Proxy Relations and Dependencies on Galaxy and Group Properties.
- East-Asia AGN Workshop, Kagoshima, Japan — 25/09/2023
— Talk. Title: Hosts and triggers of AGNs in the Local Universe.

Funding, scholarships & prizes

- Special Joint Postdoctoral Funding, China Postdoctoral Science Foundation–Anhui Joint support program, Grant No. 2025T004AH — 01/08/2025–31/07/2027
— ~22k € (allowance) + ~24k € (research grant) — competitive; ~30 awardees province-wide.
- CAS Presidential Scholarship — 04/07/2024
— ~600 € — national PhD award (480 awardees nation-wide).
- Outstanding Graduate, USTC — 15/04/2024.
- CETC-14 Guo Rui Scholarship — 01/12/2022
— ~1k € — sole USTC Astronomy awardee that year.
- China Scholarship Council (CSC) Fellowship — 20/10/2022
— ~18k € — sole USTC Astronomy awardee that year.
- Cyrus C.Y. Tang Foundation – WFST Project Grant — 01/06/2022
— ~600 €.
- Best Oral Presentation, 23rd Guo Shoujing Symposium — 17/05/2021
— ~120 € — 4 out of 180 participants.

Supervision & mentoring. Mentored MSc student (Ma, L.) on weak-lensing and satellite kinematics analyses, leading to co-authored publication (paper #3: ApJ 984, 101, 2025).

Major collaborations & networks

- UMass Amherst (Prof. Mo, H.; Prof. Yun, M.) — BCG assembly; dwarf-galaxy kinematics; VLA proposal in prep. (26/06/2024–present).
- CEA Paris-Saclay (Irfu) — weak lensing systematics & UNIONS (with Dr. Kilbinger, M.), 15/04/2023–present.
- USTC (Wang, H.; Wang, E.; Chen, Y.) — galaxy evolution; galaxy–halo connections; AGN triggering scenarios; JWST/DESI clustering (01/09/2018–present).

Technical skills

- Programming & HPC: Python; parallel/distributed computing.
- Observational cosmology: weak lensing (shear, PSF systematics), two-point correlation functions, satellite kinematics, HOD/stellar mass-halo mass relation modelling.
- Galaxy science: morphology measurements; AGN/host diagnostics; survey pipelines (image coaddition).
- Methods: Familiar with training AI models, shear processing pipeline and image coaddition.

Languages. Chinese (native); English (proficient).

References

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| • Dr. Martin Kilbinger | — CEA Paris-Saclay (Irfu) | — martin.kilbinger@cea.fr |
| • Prof. Houjun Mo | — UMass Amherst | — hjmo@umass.edu |